Biological Datasets for Computational Physics

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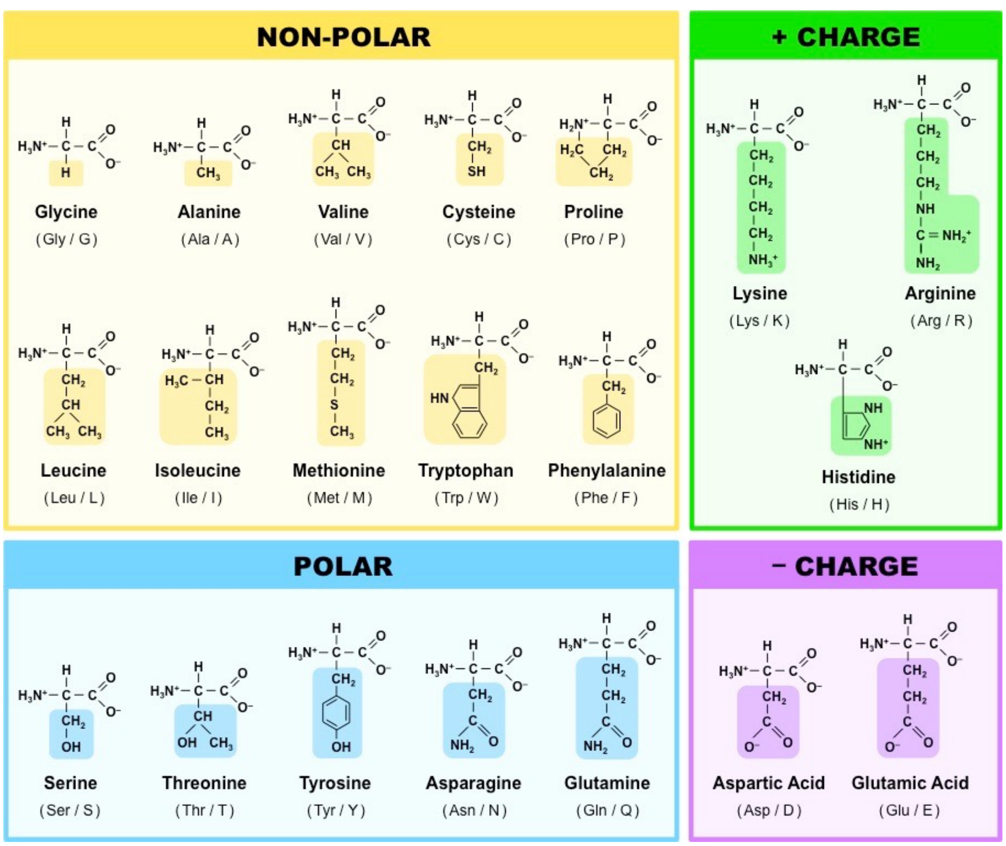

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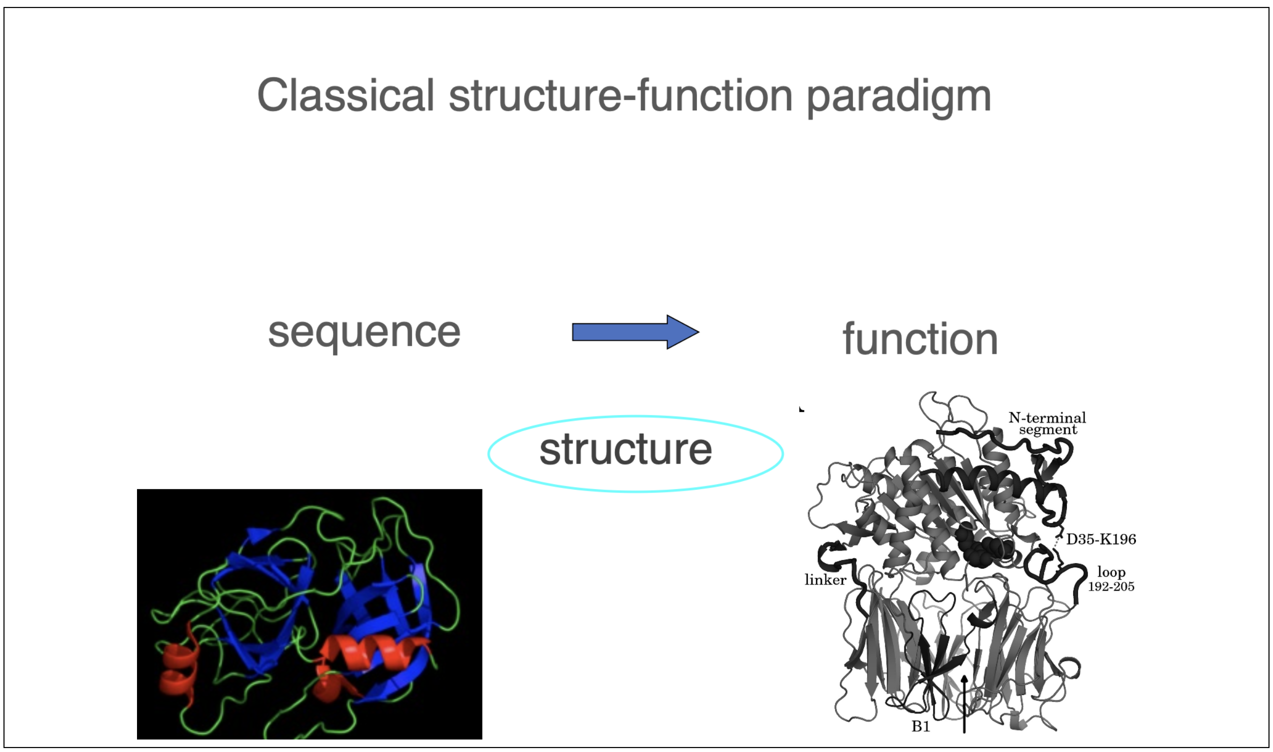
Chapter 1

Protein Structure

1.1 Protein Structure - Lecture 4

The Classical Structure-Function Paradigm



People work with this. It's called a classical structure-function paradigm, and they believe that the protein sequence encodes the function through a well-defined structure of a protein. What you need to see as a physicist is that this is a **deterministic relationship without any uncertainties**. This is what biologists believe in, this is what chemists believe in, and this had been a very strong constraint on the whole field of biology, and this is what gives you opportunity to work in the coming decades and a good salary or gain good research reputation, whatsoever. **The fact that this is not true.** This is your future. What you need to understand, again as a physicist, that this is a deterministic relationship making no notions whatsoever things that, for example, these entities, sequence, function or structure, can be **ensembles**.

Learn how to read sequences

Primary structure

MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIRLFK
GHPETLEKFDKFKHLKSEDEMKASE
DLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHA
TKHKIPVKYLEFISECIIQVLQSKH
PGDFGADAQGAMNKALELFRKDMASNYKELGFQG

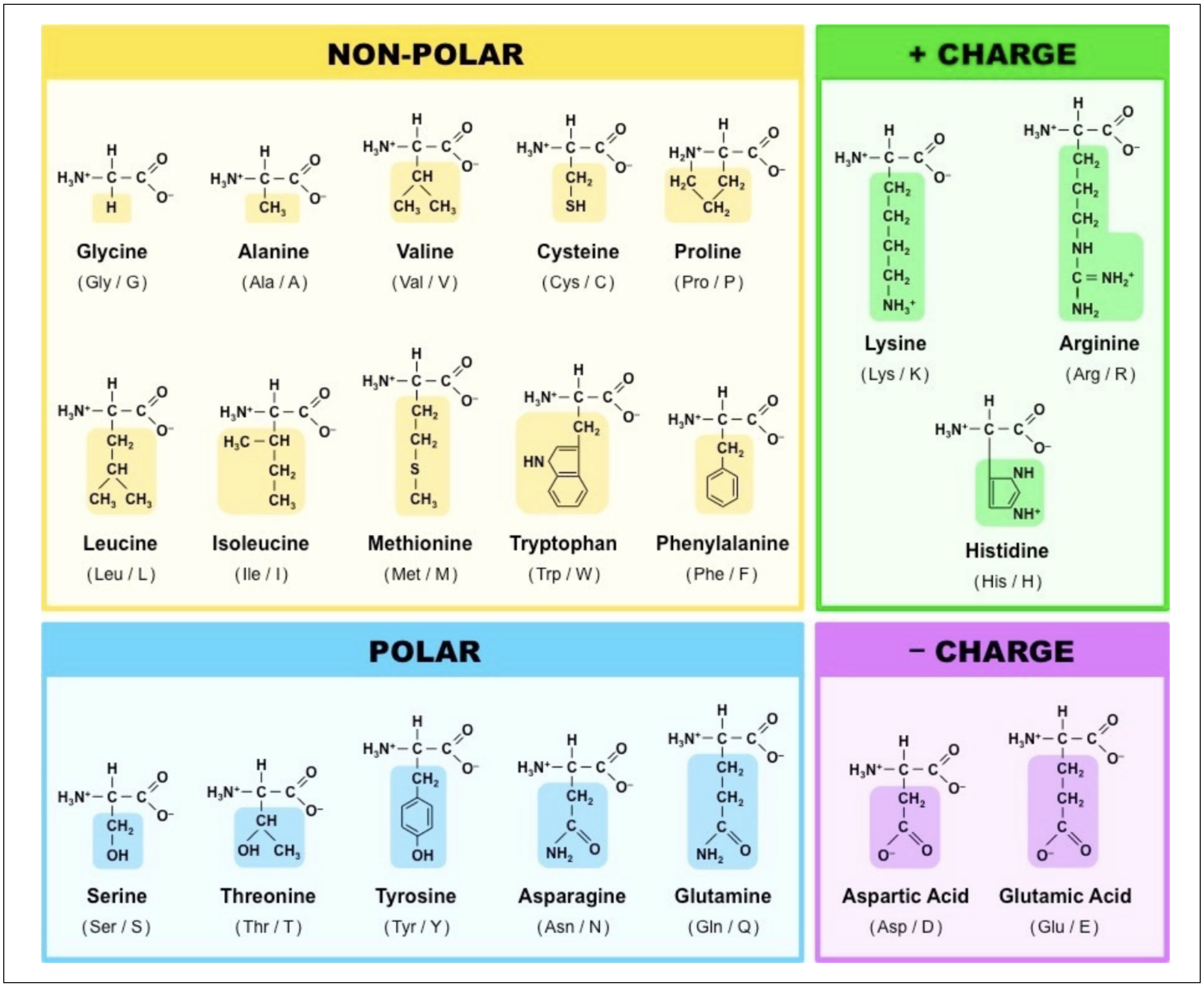
Let's start with the primary structure. **The very first thing I suggest you to do is to learn what the letters mean.** Because you need to read it. Try to read it with your, I mean, with your intuition, nd try to note something. Believe me or not, I'm working with proteins kind of every day, and I am reading these. And sometimes from my own reading understand something, because why? Because the brain does some kind of well, intelligence. So, let me help you, let me teach you to read.

If you look the whole text, do you find more or less the 20 letters? I mean, just as a first sight. Yeah, I mean, there is quite a bit of variations, maybe you don't need to there is quite a bit of variation in the letters. This we can say. Then, you know, if I start reading it, there is one follows another, so it's not the same letter. So, it's there are different letters. And then you keep reading, and then you let's say arrive at this line, you say KK and GG and KKK, even three the same. So, there are some things which are kind of repetitive.

The Aminoacids

So, we have 20. All the 20 have different properties. Now, this picture I suggest you to print out and put above your tables when if you really start doing some biological calculations.

Polar and Non-Polar You see here words non-polar and polar. This means polar ones like water, non-polar ones don't like water, they prefer oil. I will only speak about very



simple things and keep yourself simple even to do the best machine learning in the world. Okay? You don't have to be complicated, you just have to be clever.

You see that you have lots of non-polar I mean, if you just again, briefly look at the numbers, you the the largest group is the non-polar. **Why?**

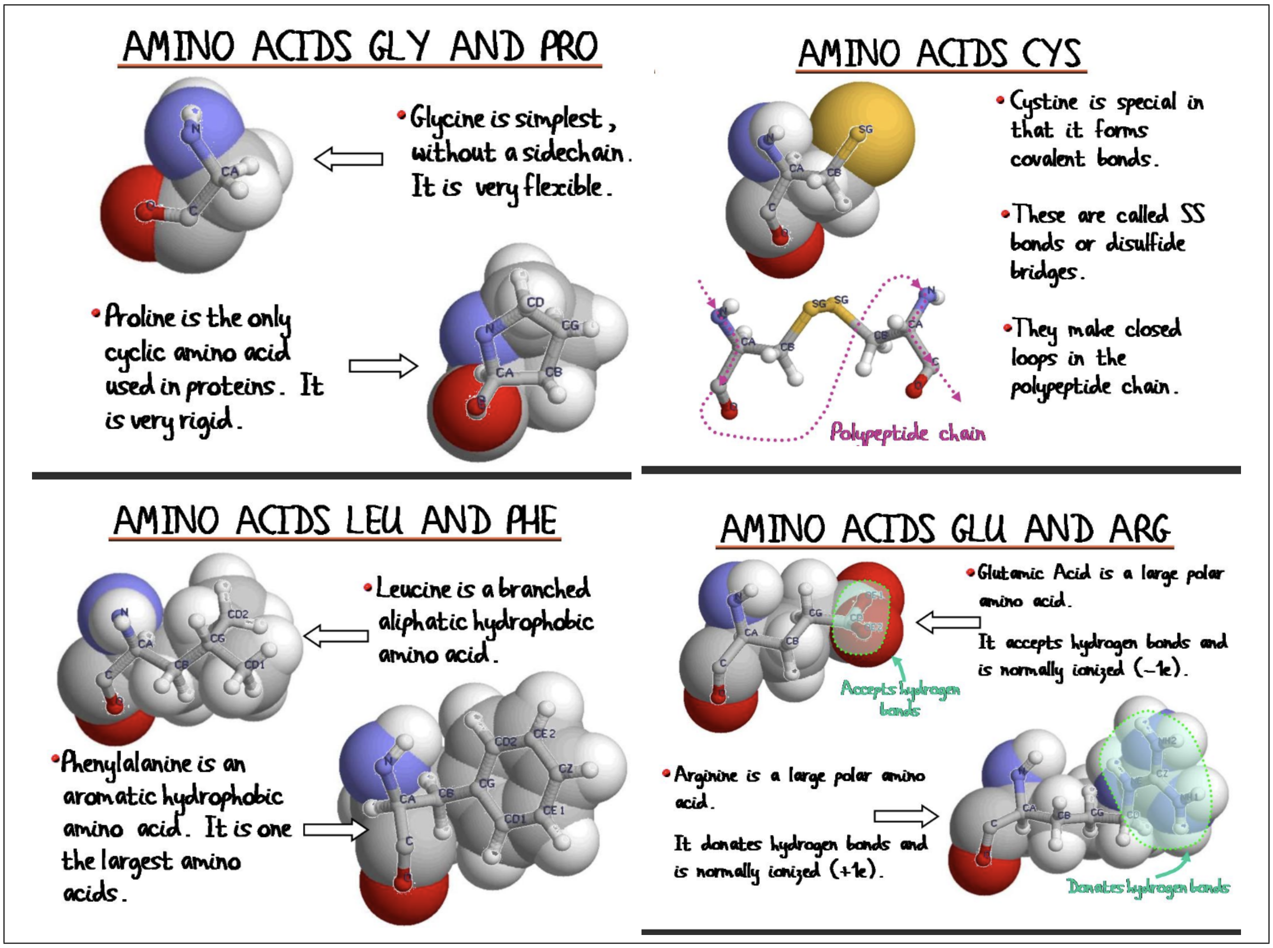
So, at the end of the day, we construct proteins that are several hundred residues, let's say. Most of them are compact. So, we have to get them compact. We construct them in a water-like environment. So, we have to make them compact. Now, if we put all the polar and later on charge, I they like water so they don't want to be compact simply.

Protein Charge depends on pH We say charged, but it depends on which solution they are or which protein they are. I mean if you put the pH very high, then, you know, everything will be deprotonated. If you put the pH very low, then everything will be protonated. So, you know, it depends. So we can say it's charged, but when you compute machine learning, you have to put one number at that pH or at that protein or at the cytosol or somewhere. So this is not an absolute quantity, this is what I'm trying to tell you.

Useful Properties of some Aminoacids

I'm trying to call your attention to some properties that you may pick up when you do the machine learning. All amino acids are very special,we have to pick up their speciality, and understand if it's relevant to the problem that we are studying.

Coure



Glicine Glicine is the simplest amino-acid. Its side chain only consists of one hydrogen atom. This has two important consequences:

- it is very mobile and flexible,
- it is not chiral

Flexibility implies that glycine is very involved in some processes, for example if you learn about transcription, transcription is a super dynamical process. The enzymes that are doing it must have some tails, some parts that are full of glycines because it must be very flexible.

Proline Then we have proline here. Now, proline is also very special. There is a covalent bond between the end of the side chain and the main chain. It is the only cyclic aminoacid that is used in proteins. Chemically actually it's called an imino-acid, it's not even an amino acid, but this question we ask only from the medical students if you want them to fail the exam. But what is important for you is that this is a big constraint: somehow it limits the conformations and basically it will restrict **proline to two main states. cis and trans** - more on that later. So things that want to change their function in a kind of binary manner would use proline because once in this state, once in the other state, only because of that constraint.

Cystine - Dsulfide bridges I take another amino acid—I won't go all through the twenty, I'm just giving you flavors. I take another which is called cysteine and cy—normally most of the side chains, if you look carefully, have a carbon, have oxygen, and nitrogen, these three. And hydrogens, of course. **This one also has a sulfur**. Big—first of all, you can see it's big, bigger than the others. Otherwise, there is another amino acid which is about the same size just there is oxygen here. Why it's interesting to have the sulfur? But the sulfur is a very sensitive atom in a sense that **it can be SH if it's reduced, and it can be SS - sulfur bound covalently to a sulfur - if the conditions are oxidizing**. Having said that, since this is, you know, on one part of the main chain, maybe, you know, the other is on a completely different part of the main chain, through this kind of bonds, which are covalent bonds again, formed only if the conditions are proper, you can put a strain. When you form a covalent bond, you make a strain. You put a strain on the whole protein, you stabilize it, and many cases you have the situation when—I mean you—you have a protein, you know, something, and you cut it, actually it's happening in your stomach for example. You cut, but it doesn't fall apart because there is this kind of bond. Before this kind of bond is formed, and then you can cut the chain.

Phenylanine (F) You can see that there is a ring here. You can see that there are six members and the particular feature of those six is that they are **planar**. There are two energetically favoured conformations of mutliple phenylanines: You can make of course a sandwich (**think about protein condensates!**), but the other is a T-conformation, this is energetically favored. This means that here you can very easily generate multiple states in a very easy manner. Because the benzene ring has a cloud of delocalized electrons above and below its plane (the π system), Phenylalanine residues like to "stick" to each other or other aromatic residues (Tyrosine, Tryptophan):

- The Sandwich (Parallel Stacking): Two rings sit flat on top of each other. While common in textbooks, this can actually be less stable due to electron repulsion between the clouds.
- The T-shape (Edge to Face): The slightly positive edge (hydrogens) of one ring points into the slightly negative face (π cloud) of the other.

Leucine You see the leucine, we say it is branched, because here there is a branch. Why it's important? **Well, when it's branched, I mean it can occupy more space, and**

there can be more water released to the bulk, this kind of things.

Glutamic Acid There is another residue which is negatively charged, which is one carbon less, so only this one and the carboxylate group comes immediately, that is called aspartate. You can ask immediately, why nature created two which are more or less the same? Not the same, because one can be protonated at a pH of four point three and the other at three point nine. This is aspartate and this is glutamate. They—it absolutely sets them for different enzymatic reactions. Nature played a lot in which cases uses glutamate, which cases uses aspartate. They many times participate in we call proton shuttling: they get a proton, get neutral, give up a proton, and so on. So that's an interesting property.

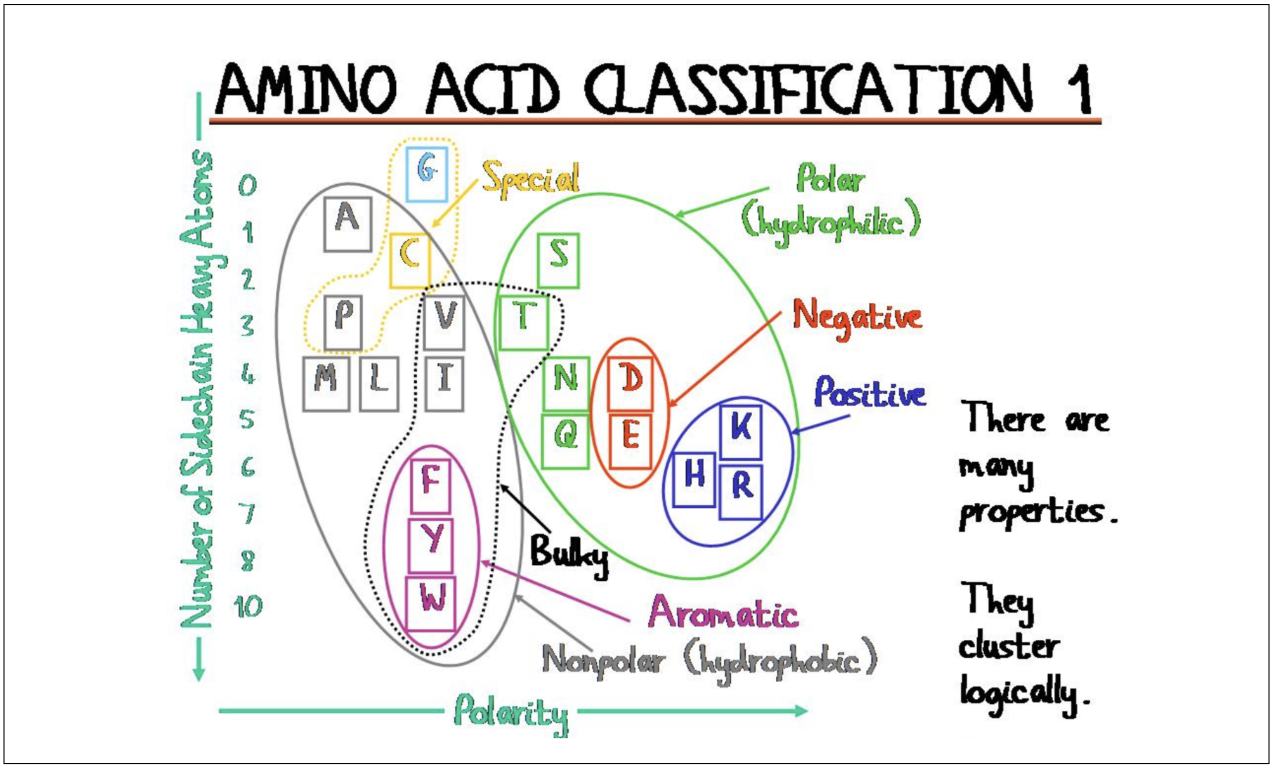
Arginine You see that there are three nitrogens here, center in the center there is a carbon, and the whole thing you may imagine as here it's—it's planar. Again, it's full of delocalized electrons. For a long time, many decades, we didn't care that much about arginine. And suddenly, like ten years ago, more or less ten years ago, we got interested in arginine. Why? Because people discovered—I mean it's—it's stupid—but people discovered that arginine has this delocalized electrons, and similarly what I explained about phenylalanine, it can make the sandwich, it can make the T-conformation. And what they realized first it was from machine learning, that this arginine is very important to form the protein condensates, this droplets in the cell. But they didn't understand why. So they had to go back to elementary chemistry. So it was first came out from machine learning really, published big journals, you know: the arginine is important.

Amino-acid Classification

- **Low polarity, many heavy atoms** So anyway, uh, so what can you expect from this group, for example, which are have a low polarity and have many heavy atoms? What—what do you expect based on the knowledge you have already, let's say? They would attract many non-polar molecules to themselves. So they would—they would have the compactness, right? You would expect that they would be inside, maybe, of this compact issue, right?

- And then you can come to here and what you expect, these are the positively charged ones, this is arginine and this is also a ring-containing compound. What do you expect? You expect either that they stay outside because they like water, or you expect that they would create a very specific environment for some reactions and so on, same as for these one.

- **Serine (S) and Threonine (T) are switches** And more you go towards kind of medium polarities, you would expect kind of ambiguity, so, you know, behave like this, behave like that, behave like this, behave like that. This kind of things nature like, to change something. So these ones, These are small residues, serine and threonine, they have a hydroxyl (OH^-) group. This group is relatively small and neutral, but it is chemically nucleophilic, meaning it's ready to react. The hidrogen atom in the hydroxyl group can be replaced by a phosphate group (PO_4^{3-}). You go from a small, uncharged,

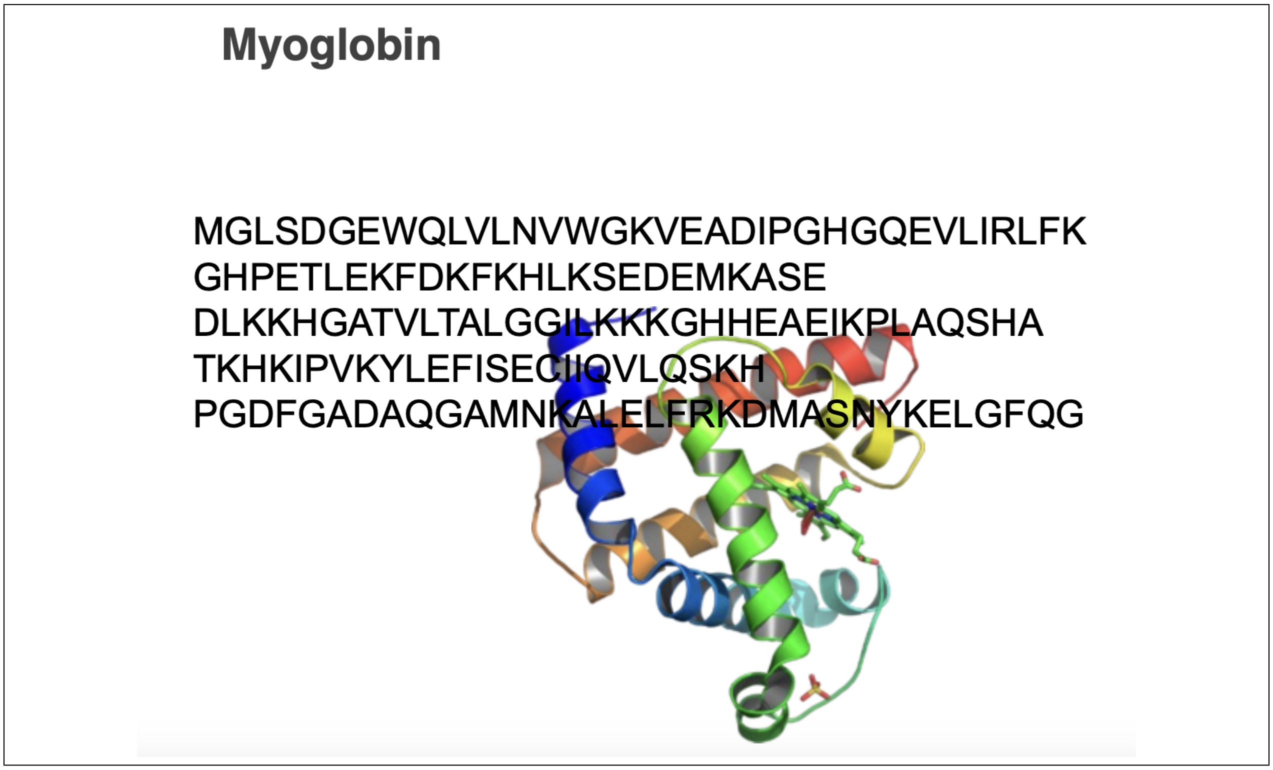


slightly polar residue to a huge, bulky, doubly negative group. This massive change in charge and size usually causes the protein to "snap" into a different shape, **turning an enzyme "ON" or "OFF."** This is the basis of nearly all cellular signaling. Nature likes regulation, it's very important for complex organisms. It's all about regulation, not about the number of genes, not about the length, not about nothing else, it's about regulation So these ones, that get this post translational modification, they change and then they make something. For example, they pass the signal, they get a phosphate and then the phosphate is taken off.

- **Asparagine (N) and Glutamine (Q)** These are the neutral versions of Aspartic Acid and Glutamic Acid: they have an amine group (NH_2) where the others have a carboxyl group. These two are very dangerous. They- they make amyloid aggregates that I explained later, it's more complicated. Basically, if they are over-represented in a protein sequence, they can zip together into indestructible fibers called **Amyloids**. This is the structural basis for diseases like Alzheimer's or Huntington's.

Sequence Complexity predicts Disorder Degree

An ordered one: Myoglobin There are quite many of these twenty, almost probably all of them are represented here. There are not many repetitions. **So we can make a conclusion- I mean, it's an early conclusion but- but try to see the- the correlations between things. That if I have many, I have quite nice**



structure, so it's- it's quite well defined, even this perpendicularities (between helices) is important, it's- it's quite well defined.

A messy one: the Mastermind Now I show you another protein. It's- it's messy basically. Try to read again and tell me something about your reading experience.

Student: There are more repetition in the letters. Repetitions involves different letters

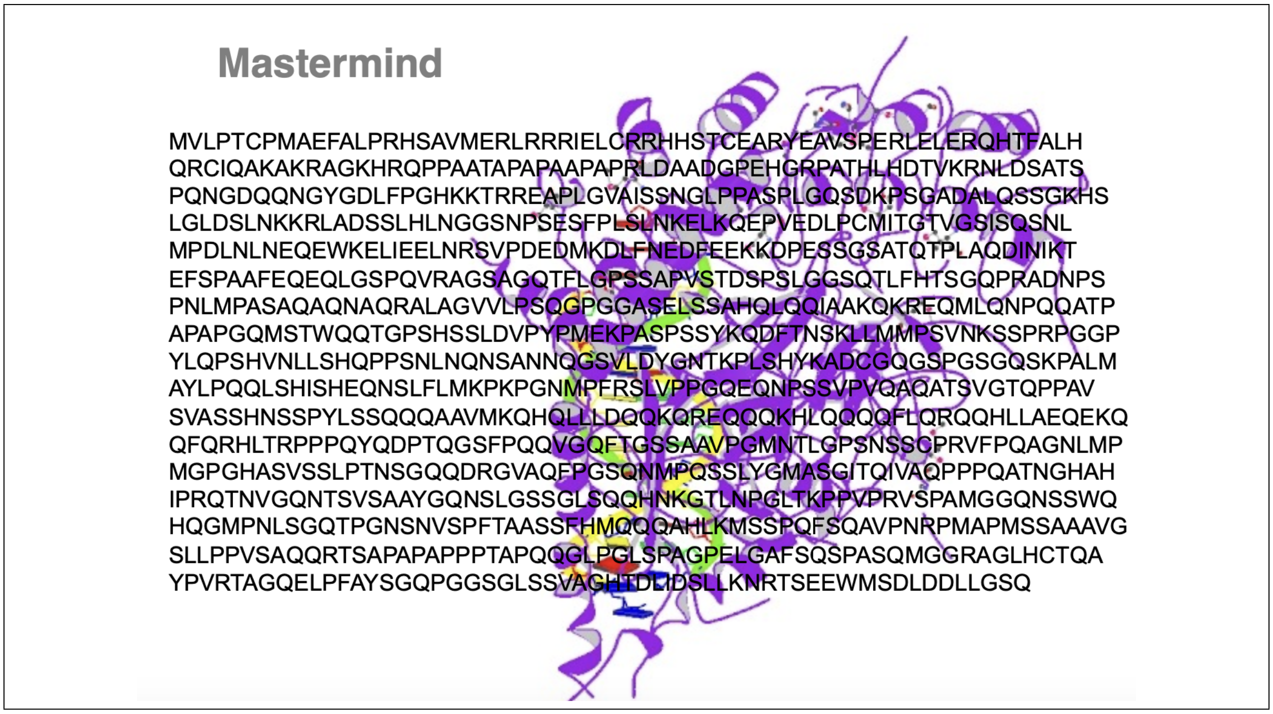
Speaker: Different but not - not any kind, right? That- there are some that are distinguished in this repetition, no? This we note. Like for example, **G,Q**, **S,S,S** . And when S is repeated, then many times there is a G there, so there is an associated G. Or if you look, another times there's an associated R.

A messy one that creates condensates One of the key proteins of the ALS disease, the Amyotrophic Lateral Sclerosis. Um, this means you have a single mutation, one amino acid changed, and it starts to grow like a hedgehog. If you don't have a single amino acid mutation, it stays normal like this.

Speaker: But read the letters. Read the letters. What do you notice?

Student: There are a huge repetitions.

Speaker: What kind of? G,G,G and what else? Q, S... if you look very carefully in the region which I highlighted, if you count how many types of residues are present or



dominant, it's different than Myoglobin. I don't know, there are maybe five residues here out of the twenty. So not all the twenty are represented. This is very important.

High-complexity and Low-Complexity Proteins have very different roles

$$H(\text{sequence}) = -\sum_{\text{Amino- Acids (AA)}} f_i(AA) \cdot \log_2 f_i(AA) \quad \textit{Shannon Entropy of a sequence} \tag{1.1}$$

- high shannon entropy = high variety of aminoacids. These are called high complexity sequences.
- low shannon entropy = low variety of amino-acids. These are called low complexity sequences.

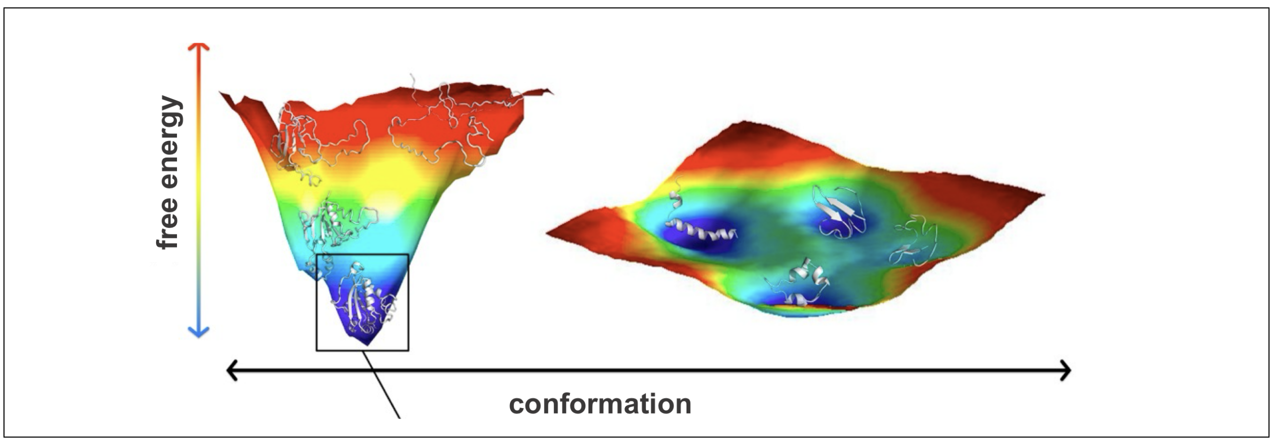
So, does it matter? Of course it does matter, otherwise I wouldn't speak about it.

- **High complexity proteins: the "Swiss watches"** This is the hallmark of enzymes and folded proteins. To catalyze a reaction, you need a highly specific 3D pocket. You might need a Histidine to grab a proton, a Serine to attack a bond, and a Leucine to hold the substrate in place. These residues must be positioned with sub-angstrom precision. If you change even one amino acid (lowering the complexity/variety), the "machine" breaks.

- **Low complexity proteins: morphogenesis** When H(sequence) is low, the protein is repetitive (e.g., Gln-Gln-Gln or Gly-Ser-Gly-Ser). These are often Intrinsically Disordered Regions (IDRs). If you are coordinating 1,000 processes at once (like during embryo development or "morphogenesis"), you cannot have $1,000$ rigid machines clashing into each other. Low-complexity sequences act more like liquid or velcro. They allow for multivalent interactions: they can bind to many different partners weakly rather than one partner strongly (**fuzzy interactions**).

A sequence could have both low complexity and high complexity domains Speaker: So how do you know that this is going to form a droplet and then ages to this hedgehog and then form aggregate? So you want to distinguish these regions. One that will be structured and the other that will not be so structured, or structured only in a case of a pathology. But for example, if you were computing sequence complexities, you should find a great difference in between these two.

1.2 The Conformational Landscape (Lecture 5)



The folding process is driven by thermodynamics laws: the system (the protein) will spontaneously evolve towards a configuration that corresponds to a (local) minimum of the free energy. In this context, the free energy that we need to consider is Gibbs free energy

$$G(T, P, N, \text{other parameters}) = U(S*, V*, N, \text{other parameters}) + PV - TS = H - TS$$

Spontaneous transformations satisfy $\Delta G = \Delta H - T \cdot \Delta S < 0$. In general, the Gibbs free energy defines a very complex hypersurface, which we call a *conformational space*. So, this complexity, let's say, faces a protein with a challenge that you will feel immediately once you push the first "start" button into your simulations, because you will see that it doesn't want to converge, there are many bottlenecks at some points that you just cannot pass.

Sometimes it is a real challenge, for a protein, to reach an **adequate state**, meaning a state where it can function properly. Notice that I used the word "adequate" - not **"native"** - as you find in the literature, to avoid two common misconceptions:

1. the state where the protein can function is unique (FALSE)
2. this state corresponds to the global minimum of the free energy (FALSE)

These false beliefs have been around in the field for decades, and it is better for you to learn the reality from the very beginning. People made some early on experiments trying to heat up proteins a little bit and then let them cool. And what they experienced, that they always get back to the same state, and this is why they thought, "Oh, there is just one of these states and the protein must find it." And for example, just to tell you why you must be cautious with this notion, that for example in your cells, in your body, the most stable state for proteins is an **amyloid state**, which is in most cases not functional but pathologic. And it is also irreversible. So, that's kind of a dead end for a protein.

The landscape is not a flat golf course (oh, really!?) At first, people had this naif idea that the native state was unique and that the energy landscape was kind of flat, with a very sharp dipping point where the native state was. But if things where like that, then doing very simple calculations, you can see that the average time for the folding process to complete would be large than the age of the universe! Instead, proteins fold in the range of **milliseconds - seconds**. So this must be clearly false.

The Funnel idea So one of the first answers were coming '87 by Ken Dill, who is still alive and active, and he proposed the funnel. *The funnel means that there is a vast space in the beginning and after a short amount of time, it's not only time but it's also the number of the contacts, the protein reaches a state from which there is only one way to achieve the final structure.* So there are multiple pathways here in the beginning and then there is a funnel. By the way, all the AlphaFold is compatible with the funnels, but not with the ensembles. Now there is an AlphaFold multimer, but they are still working on it. I'm stressing it, that would have meant that for a while you have many configurations, but after a while everything is kind of determined for a single state. Okay? Please note—I'm showing quasi each lesson on this structure-to-function paradigm, deterministic—please note that this is like a rephrased form of that that paradigm. It's absolutely deterministic.

The reality: frustration Twenty-five years ago, people started realizing that some proteins have a pletora of metastable states. They thought it was a special class of proteins, but in fact all the proteins are like this, some more and some less. The landscapes range in a continuum, some are quite sharp, some others are very shallow. And then people started to talk about the spin glass effect in proteins, okay? Obviously this community of physicists. So, what is the spin glass effect? That you you there is frustration, you cannot fulfill all the interactions in a in an optimal way, but you must leave some kind of frustrated, unfulfilled. And these states cause little bit of variation of something that we would call "ideal" or "native". You should not say "that's non-native, so that's not functional" - it is

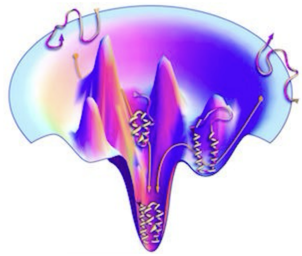


Figure 1.1: the funnel model

not true. These are marginally stable states here, and you have some conflict in energy. IT doesn't mean that it's non-native; **it means that it can be native under some different circumstances**. It means that when evolution comes and let's say faces the protein with with something different than it got used to—for example, the temperature gets higher, for example happening now, right? Or the CO2 concentration becomes higher, or thousands of other things, okay?—then the protein can shift towards this. These states are very im-

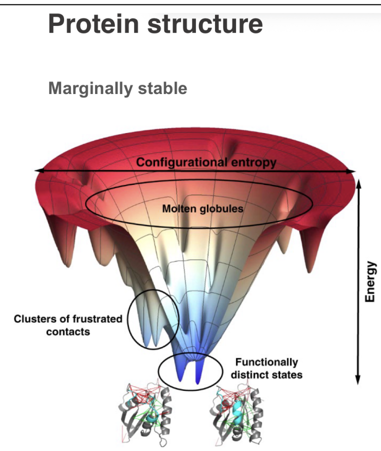
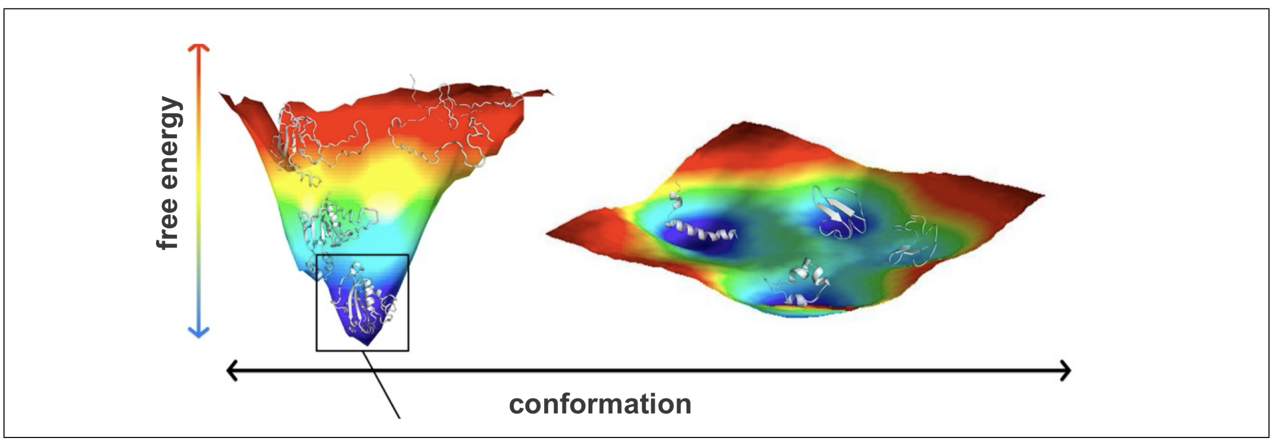


Figure 1.2: Frustration

portant. So, the fact that you don't die every day from eating the food that you buy in the supermarket which contain chemicals from pesticides that are absolutely new on earth, like they were discovered 10 years ago whatever. All your proteins were not evolved to deal with this, it's absolutely new, it's absolutely dangerous for you. But your proteins are very intelligent, they have this state, and because they have this state, this state can degrade this unknown substances, they adapt, and well, you are here, you are listening to me. **You need to understand that what we call "non-native" is just something that is not the major function, but the protein is evolved to have this kind of secondary activitie**s, moonlight activities, whatever, through the frustration of some of the contacts. Tt's not random which ones, okay? It's very important. It would be fantastic to understand which ones they to be, but definitely there must be this kind of small, let's say, variations, ruggedness, **in order to survive, in order to adapt**.

Student: So uh what you're saying is that this high degeneracy of local minima is actually a feature of proteins which helps it to uh be flexible to in different situations? Am I right?

Professor: To be adaptable. Exactly. This is what I'm stressing. I mean most people—I must say this because you may read papers related to your problems, related to your projects, whatever—most proteins are imagined to have one unique state, one unique functionality. But as you are physicists, you know that there is a conflict in between, there is a degeneracy, and there is a dependence on the environment. So, it's absolutely an elementary essential property of proteins, and I try to teach you from the beginning with the right theory, let's say, that if you have a good imagination, I think it will be easier later on to do any kind of computation. So you can, you know, you can play and nature plays. That's the very important thing. Nature plays because we must be adaptable. We, including us as an organism, including us as cells, including us as individual proteins within our cell. And this is amazing thing. This is amazing thing and I think because you are physicists and you can truly understand the valid nature of these landscapes, you can appreciate the complexity coming from this little noise — rather I don't know shouldn't be noise it's essential for our life, but this little variations that there are. Without those we would not be alive. And that's fantastic how to create something which is optimized largely, but just largely - not fully.. It's a miracle really. It's just right.



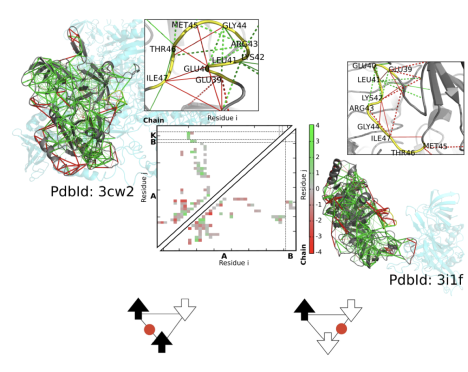
Intuitions: what makes the landscape more or less shallow You can compute the average amino-acid composition of proteins across the entire human protein database. You will find that disordered proteins show interesting variations in their composition with respect to the global average.

The aromatics (you recognize phenylalanine, tyrosine and the double ring the tryptophan) and the small hydrophobic here they are all depleted which means that they don't have sufficient amount of these hydrophobic residues to form the core. Also, you can see in particular that the big ones (the "bulky" amino acids) are missing. And in contrast they are enriched in the charged ones K, R the positive ones E and D these are the negatively charged ones and there are some others that you may recognize from the last week like glycine—which is a very flippy floppy residues because it doesn't really have a sidechain. You should expect proteins with these characteristics to have a shallow energy landscape.

Indeed, these characteristics mean that they don't want to adopt one structure, or just few structures. These proteins don't want to have a few - they want many. Because the charged amino acids are like water. They don't want to come together. And they are depleted in hydrophobic residues that would provide this hydrophobic effect.

Functionality regime: change a little, not too much Well, as I said, evolution will be a major argument today. So, what you need to understand is as follows. For a protein to function, there is a regime. Of course, when the protein looses too much stability, it looses its functionality. But the problem is that there are some aspects of the function—for example, dynamics, some conformational changes of regulation—that cannot be achieved if the protein is too stable. So, there must be a regime, okay, where the protein is kept for that particular functionality. And at that regime, the protein can accept many mutations which doesn't really change its whole organization but keeps it rather adaptable depending how the conditions are changing.

Some proteins are extremely mobile In some classes of protein—all proteins move—but in some let's say functional classes, okay, they move extreme. And this is when the conformational energy landscape becomes rather shallow. For example, this is a homeodomain. Homeodomains regulate gene expressions and they are very small, they are super mobile, and they they control absolutely crucial processes by some small bits that you don't even see here because you they couldn't determine the structure; it's just too mobile. But these tails determine something about DNA specificity, whatever. And if you—if you trigger them, if you, you know, play around with that sequence, uh there is this famous experiment that you can grow a leg from the fly's head, okay? So, you can let's say generate this kind of creatures with very small modifications in the super-super mobile part of the of these types of homeodomains.



Difference is enough for specificity? I was lucky enough a few years ago working with a pioneer of this theory, and what we tried to explain that, okay, if proteins are frustrated, if they are not optimal, then how come they are specific? And we computed this: this is a same protein with two different substrates. And anyway, what you see here on the two triangles, these are contact maps with two different substrates. And what you see also, it's full of red, it's full of frustration, so it's not optimal at all. But you can even see by eye—but obviously we can compute—that the pattern of the red dots are different. So, it's not optimal, not optimal at all, but it's different. **And difference can be enough for specificity.** So, we have not finished this study yet; if anyone is interested, it's open.

Surrounding environment can change the shape of the energy landscape Student 2: I have a question. Once we have a single protein—so it's amino acid chain—is the energy landscape fixed or for example some changes in the environment can change it?

Professor: I understand, you are asking such a difficult question. It's a very good question, a very, very difficult question. So, what you typically do is that you write your potential as a sum of terms, each accounting for some interaction - and then you add at the end a factor that is coming from the environment.

$$U = f_1 + f_2 + \cdots + \zeta$$

So maybe interaction with a solvent, interaction with an ion, interaction with something. And then things change, yes.

1.2.1 Forces that drive the folding process

So, first we speak a few words about—about the organization, about thermodynamics, okay, before touching the question of the folding itself. So, obviously we are somehow optimizing, but again not super-optimizing the free energy, okay? So, there are terms that contribute to the enthalpy and there are terms that contribute to the entropy.

Interactions So, one of the key things is hydrogen bonding, okay? This is what keeps proteins kind of stable. Although the overall stability of a protein—let's say a protein of 300 residues—the overall stability of a protein is around like maybe five hydrogen bonds, the energy of five hydrogen bonds. It's marginal. We say it's marginal. Hydrogen bonds normally means that there is a—there is a donor, there is an acceptor, and this bond is usually shorter—it's longer than a usual covalent bond, but but is normally shorter than a usual secondary or non-nonbonding interactions. Normally the distance between the heavy atoms is between 2.5 to 3.5 Angstroms, and this is a very—this is a very interesting bond, we'll speak more about it. This is something crucial. Uh, you had seen the amino acids, some of them are capable of of forming these.

Then we have covalent bonds, okay? You remember cysteine, and uh upon certain bio—environmental change, at certain electro—let's say redox environment, it can form this covalent bond, okay? And I explained to you that this is good because you can even cut—I mean, the protein is kind of a continuous chain or a set of continuous chains, but if you have these bonds you can even make a scission of some parts of the protein and this will keep it intact.

Then we have kind of also non-bonding interactions, and we can say it's separated here, hydrophobic interactions or Van der Waals interactions. There are interesting forms of how we express this, okay? Uh, with the Lennard-Jones potential, Morse potential, we'll discuss it later, but there are some attraction coming from the hydrophobic groups.

And obviously we do have electrostatic interactions.

Constraints on chain rotations And uh when I start rotating them, depending what is at the position of R—so which side chains I have—I can have some some bumps, right? If it's too big and I start turning it, then they bump one another and then you can draw basically very limited area along Phi and Psi where actually these transitions are reasonably smooth. So, this chain geometry put some constraint on the protein. There are some regions where it's kind of allowed and there are some other regions where it's not really allowed.

Then there are some other terms that again we will talk about when we construct our potentials. Things that are like two-body interactions—these are bonds, okay?—then angles, so some parameters of the actual molecularity—these are absolutely like chemical terms, okay, like angles. For example, this is a planar ring, a phenylalanine ring, but obviously if I start let's say whatever a little bit tilting it, it's unfavorable, okay?

Or there can be relations between 1-to-4—these are about we call torsion angles. And it's not—it's a small term but it's a very important term because we have many 1-to-4 interactions and in many cases you you must make sure that the 1-to-4—so these are two atoms that are... Sorry. So, you have two atoms that are covalently bound to one another, 1 and 2, and you have two others, let's say 3 and 4, that are not covalently bound but obviously their position are largely constrained by the fact that you have two atoms bound. And you can rotate 3 versus 4—so you can start rotating it—but for example if this is a big group, when you rotate to here, there can be, you know, some overlap, some unfavorable repulsion. So, the orientation of 3 and 4 is kind of—I'm not saying it's fixed, but there is a certain kind of energy function associated with it. And this is quite important, this is what called torsion.

Okay. So, this is kind of the geometry of the actual residue, okay? And then you build up on it the geometry of the chain, which is some kind of local parameter.

Okay. So, basically all these, let's say, three major areas would determine something that people associate with thermodynamic stability of the three-dimensional organization.

But then comes entropy, which most people don't really think about, okay, or think about in a very limited way. So, what contributes to chain stability is obviously the mobility of the whole chain. And here you see a structure colored by so-called B-factor, which shows the thermal vibration of each atom in a crystallographic structure. Very soon, next week, you will learn how to make such kind of pictures and how to interpret such kind of pictures. This is rhodopsin, which is a key protein for light sensing. And you see—not so surprisingly—that the inner helices are kind of more rigid and the surfaces—but not everywhere—they are quite mobile. Uh, sorry, this is a pancreatic lipase. So, anyway, there is some kind of entropy term with this mobilities.

The Hydrophobic collapse However, the major major entropic driving force for making the three-dimensional organization is something that we kind of touched upon last week. You remember when we discussed why we have the majority of amino acids non-polar? Yeah, right? I asked this question and I got a fairly good answer for that. And I told you that this is the so-called hydrophobic effect, which means that when you have such a hydrophobic chain, water molecules are pretty much constrained, okay, if they need to contact. But whenever you can put these chains together, the water leaks to the bulk and entropy increases.

Now this term is very difficult to capture. We will speak about it because I mean once you not speculate but once you try to estimate, calculate the stability of a protein, this this is very important. However, when people determine a protein structure, this this has already been completed, this process, so all those waters are outside. So, it's very difficult in fact to trace how much water molecules have left, how tightly they were bound, and what is the entropy gain. Normally the gain from the entropy, even if you take a normal protein, okay, not this kind of extreme disordered protein, is comparable to the interaction energies that you that you get from all these kinds of enthalpy terms. Okay?

Marginal Stability So, at this moment, before being able to run a simulation, you just need to accept, again, that whenever you sum up all the terms for the protein stability, the ultimate value that you get is around between 5 to 10 kcal/mol. This is about the enthalpy of like five hydrogen bonds.

$$\Delta G = G_{folded} - G_{unfolded} \simeq -10 \quad \text{to} -5 \quad \text{kcal/mol}$$

To put that in perspective: a single covalent C-C bond is about 80 kcal/mol. The entire $3D$ structure of a protein is only as stable as two to five hydrogen bonds! If you heat a protein up by just a few degrees (like a high fever), or change the pH slightly, that $5-10$ kcal/mol stability vanishes, and the protein "melts" (denatures). Why would nature design something so fragile?

- Flexibility and Function: If a protein were as stable as a diamond, it would be a "rock." It couldn't move.
- Protein Turnover: The cell needs to be able to degrade proteins when they are no longer needed. If they were too stable, the cell would become a "junkyard" of old, indestructible machines.
- Regulation: As your notes mentioned earlier, small changes (like adding a single Phosphate group) need to be enough to flip the protein's shape. This is only possible if the protein is sitting right on the edge of stability.

Modeling the hydrogen bonding A hydrogen bond is an electrostatic interaction between i) a hydrogen atom that is covalently bound to an electronegative atom (like O, N) - thus having a partial positive charge, and another electronegative atom with a partial negative charge. Hydrogen bonds help stabilize the protein structure. They are key in α-helices. Unlike a purely ionic bond, which pulls in all directions like a magnet, an H - bond is directional. It is strongest when the three atoms (Donor - Hydrogen ... Acceptor) are in a straight line. When we look at protein structures, we don't just look for distance; we check the angle. If the angle is too skewed, the H-bond is weak or non existent. So initially people thought "Oh, I need to construct a term for the hydrophobic interaction, I need to

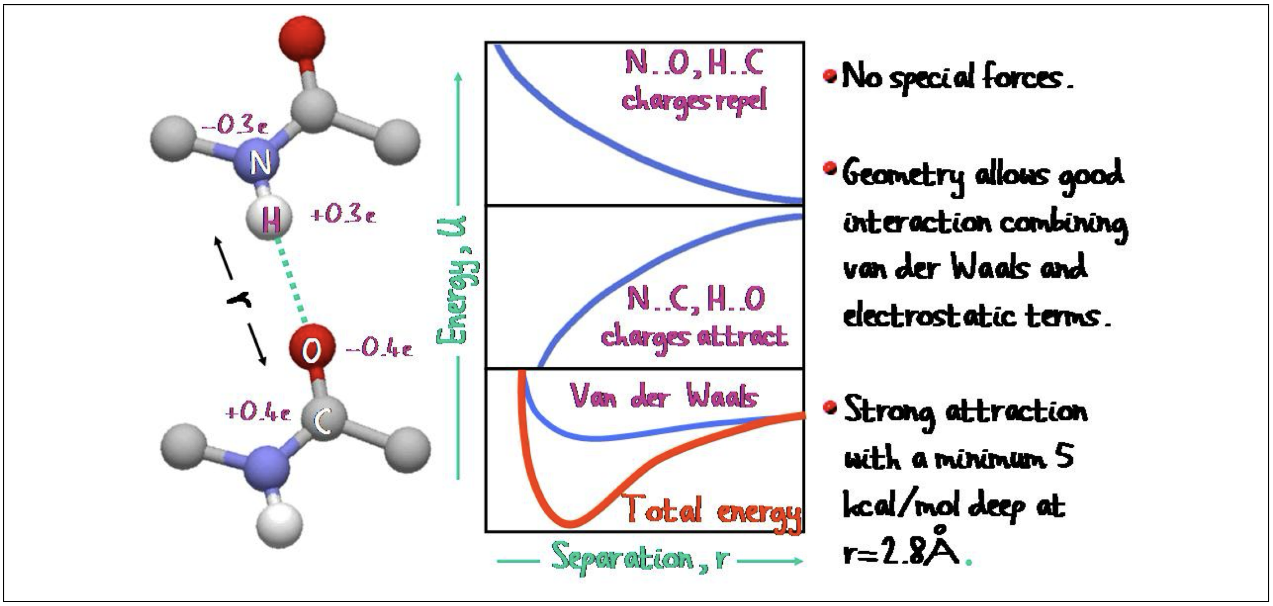
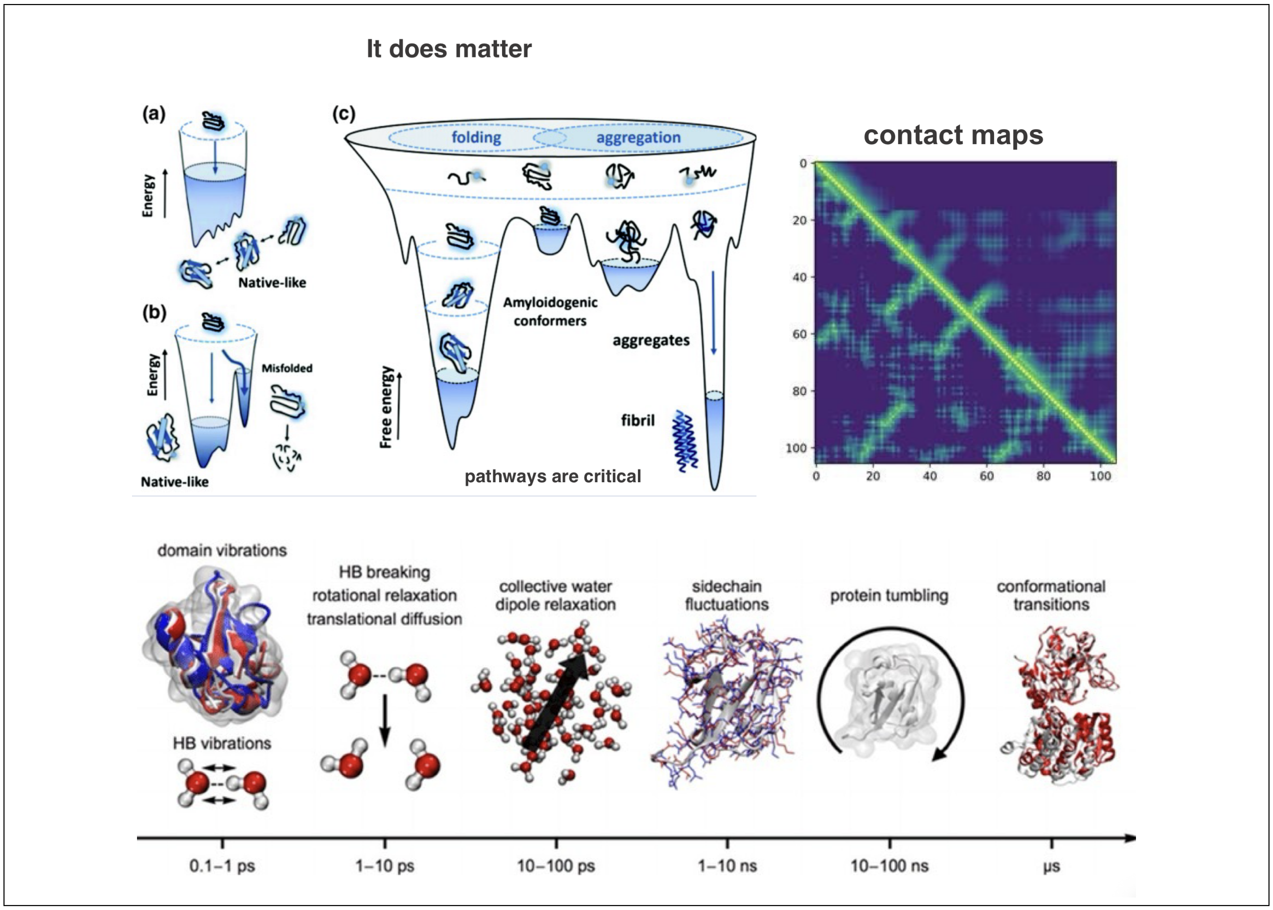


Figure 1.3: Caption

construct a term for the electrostatic interaction and plus I need to do something for the hydrogen bond" . It had been a tremendous landmark that you don't need these terms at all. You just need to consider electrostatic interactions and Van Der Waals interactions, and the hydrogen bonds will reveal themselves as emergent from those. This was around the 70s, the heroic era of biophysics, and it lead to a Nobel prize.

The pathway determines the final folded state There are different time scales as you see here. You need to consider all these motions. But at the end of the day the protein can reach that state within the time scale that is meaningful for our life, meaning that it around millisecond everything is completed. Okay? But these things cannot occur completely simultaneously. Some of those are obligatory, you know, in a consecutive manner. So it's a huge problem actually. And without correct folding... well we will mention what happens. Now the question is, does the path matter? It does matter.

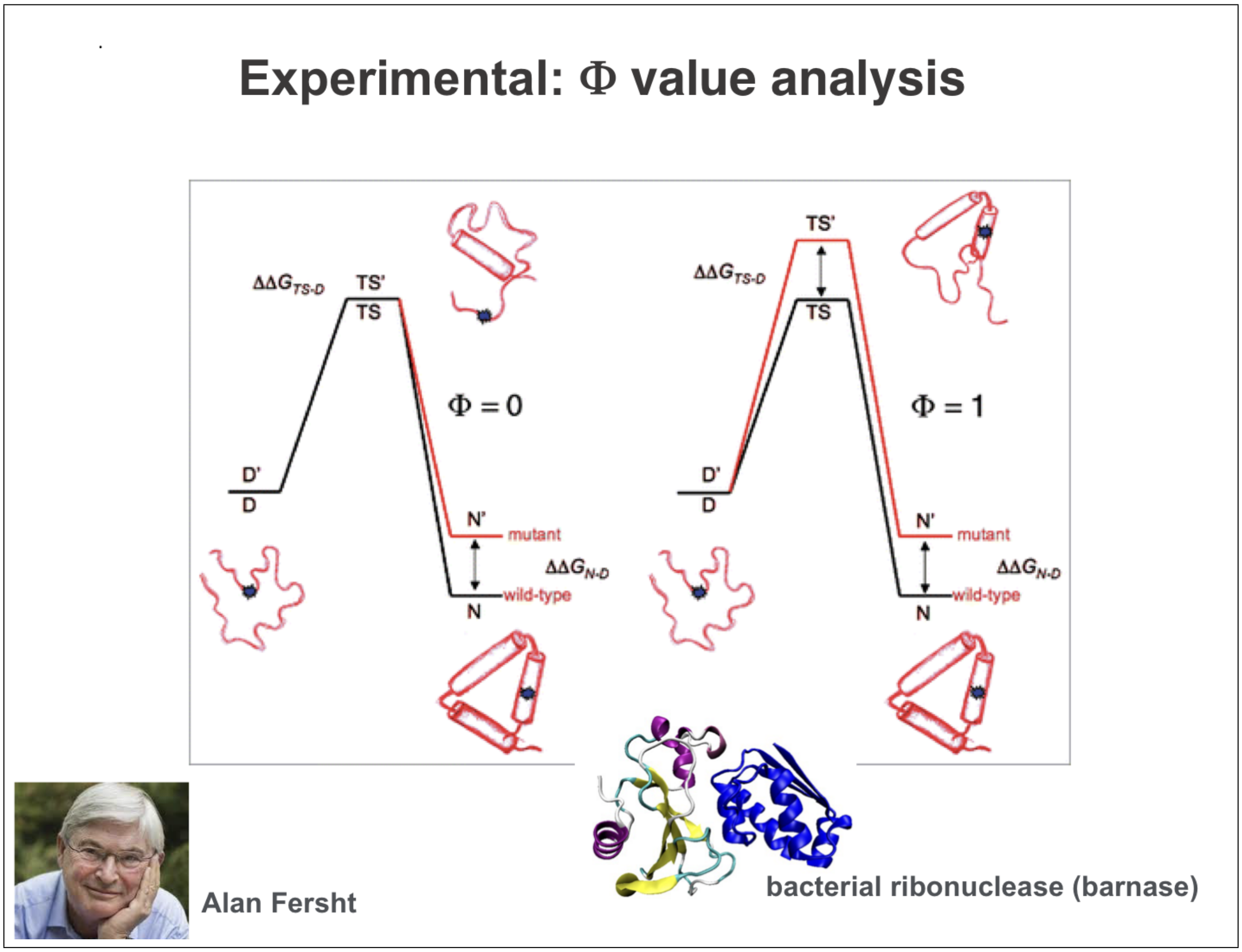
Folding stages exist but are not so cleanly distinguishable So folding qualitatively can occur in many different ways. Initially people thought 'Oh, proteins can form helices, beta strands, they are formed and then they come together' with diffusion and collision. However, because of frustration the global organization sometimes overrides whatever is happening at the level of the secondary structures. So maybe if I cut this protein at this point, I would have this nice helix. However when I fold up everything completely, it's possible that I'm losing this helix completely because the local contact, the local hydrogen bonds may be less favorable than let's say the contacts of this one with some turns or whatever. But to some extent the secondary structures are preserved. Interestingly, for those proteins that have this very shallow landscape, the secondary structures are better preserved than those proteins that are very nicely packed. This is something that you can only prove by calculation.



Contact maps A protein contact map represents the distance between all possible amino acid residue pairs of a three-dimensional protein structure using a binary two-dimensional matrix. For two residues i and j the ij element of the matrix is 1 if the two residues are closer than a predetermined threshold, and 0 otherwise. Various contact definitions have been proposed: The distance between the Cα-Cα atom with threshold 6-12 $\mathring{A}$; distance between Cβ-Cβ atoms with threshold 6-12 $\mathring{A}$ (Cα is used for Glycine); and distance between the side-chain centers of mass. Contact maps provide a more reduced representation of a protein structure than its full 3D atomic coordinates. The advantage is that contact maps are invariant to rotations and translations. They are more easily predicted by machine learning methods. In the final protein if you make a catalog of all the interactions that you observe. Basically you can generate some kind of a contact map. If you have no fixed catalog, but instead you have a fuzzy, changing contact map, then it means that the protein is disordered.

Hydrodynamic radius is a possible measure of the advancement state of the folding process. It is a measure of how much space a protein "takes up" as it moves through a liquid (like the cytosol or water). It is not just the physical size of the atoms; it includes the layer of water molecules that stick to the protein's surface, because when a protein is in water, polar groups attract a layer of water molecules (the solvation shell). When the protein moves, that shell moves with it, like a "water jacket".

Φ value analysis Well mostly I will try to say something from the point of view of theory, but I also mention that it's possible to approach some bits experimentally.



D is the energy of the denatured state, TS that of the transition state, and N, that of the native state. The transition from the denatured phase to the final folded state is represented with its energy barrier.

The folding can be divided roughly into two stages, which timescales are well separated:

1. Unfolded State to Molten globule: when the protein is born, it is just a random coil. A residue (like #2) might touch residue #53 one microsecond and #91 the next. There is no "catalog" of contacts. The Molten globule is the halfway point. It is compact (the hydrodynamic radius is small), meaning the hydrophobic core has collapsed to hide from water. But the "side-chain packing" is still fluid. The "catalog" isn't fixed yet. This transition happens extremely fast ($0.1\mu s$) because it is a "downhill" process with almost no energy barrier.

2. Molten Globule to final state. And then from a molten globule to reach a kind of compact ordered state when you fix, okay, you go through an activation energy because some parts you need to expose and other parts you need to kind of repack. And

this comes with an activation energy so this is much slower. This is three orders of magnitude slower. This is $\sim 1\,ms$.

So therefore, since this is slow you can capture something here and basically what people do experimentally that they make a specific change, a specific mutation in a protein and see how this barrier changes, how the time basically changes, okay, through this transition state. And if they make one change this more or less tells the involvement of that region in forming the final structure. Okay? And basically if you do a systematic study you can do systematic protein mutations—it's called phi value analysis—basically you can tell whether a certain contact was before the transition state or after the transition state.

$$\Phi = \frac{(\Delta G_W^{TS \to D} - \Delta G_M^{TS \to D})}{(\Delta G_W^{N \to D} - \Delta G_M^{N \to D})} = \frac{\Delta\Delta G^{TS \to D}}{\Delta\Delta G^{N \to D}}$$

Because those ones that affect, they were before, those ones that don't affect, they were after. That's so simple. But it turned out basically that when you start with this very wide region, very few contacts, less than 20percent of the contacts are formed and they already kind of fix the protein towards the final—final state. This is something very interesting.

The p53 protein is related to half of human cancers So for example we talked a little bit about p53 I think before and maybe we will have one simple practical related to it. So this is for example a representative of this class (disordered proteins) So normally when you have these shallow things, okay, they not only let's say form conformational ensembles they are ideal for all kinds of regulations because you can very easily shift these ensembles. And unfortunately the *binding promiscuity* causes disease. For example I mentioned to you that the mutations of p53 are involved in about half of the human cancer. But for the moment what you need to understand that this kind of protein dynamics can get a little bit extreme, okay? Pushing or maybe washing away the boundaries of the deterministic relationships putting it more to the stochastic. Okay?

Chapter 2

Practicals

2.1 Tutorial: Protein Data Bank (PDB)

Structural Information of Proteins Speaker: So... format. Okay. So what we do today is to derive this data in a way that you, you can handle it, you can program it, okay? And you can understand some basics. So this is your goal: you want to know the atoms, you want to know all the coordinates, you want to know all the coordinates precisely. This is a huge challenge. I will mention some of the methods that can be used for this. There is a detailed information file, "Structure Determination of Proteins," and... well, later on if you wish we can, we can speak about that. They may just try to handle the data, which is basically a result of these structure determination methods.

So obviously you want to, you want to use labels, identifiers, numbers, chemical identities, residue numbers... so in PDF, which is about the biomolecular structure, I basically... I summarize your mission. Okay? So you want to deal with all these identities, you want to know the residues for some kind of analysis, but for others you really need to know the identity of the atoms, you want to know what they are doing in a reaction. And you also want to know, for example, how many covalent chains you find in your molecule. So you want really kind of a very systematic, very categorized way of handling all this data. In addition, you may find some others, but the crucial ones are basically summarized on this slide.

So afterwards your task is just to briefly overview, to somehow understand this information, which you mostly want to do with a computer, with computational ways, right? Everything just computed using equations and so on. But... I would like to point out that some of the computations they require—and you will realize that it's kind of essential—that you really see the molecule. You really see some parts and you really get some kind of intuition that later you put into the computer.

The Protein Data Bank (PDB)

Okay, so let's start with deriving structural information, okay? So for that matter, we go to the PDB. Okay, and please open up your laptops and then try to follow what I'm doing. Everything that we are doing today is also summarized in a, in a file which is like "Protein Structure Data," okay? It's a PDF file posted on Moodle, so if you want to do a recap at home you can do this, but I recommend that we do this together.

Okay, so basically that's kind of the database that contains most of the structural information of proteins. And having said that, if someone tries to publish anything related to protein structure, it's obligatory—even before the publication is accepted—to deposit all the data, and not only the coordinates that we will speak about and we will utilize, but also all the experimental parameters, all the experimental datasets, so that they can be controlled.

Methodologies and Statistics

You can see some statistics here... and in fact I will open up some, some lists here because we will navigate from here, but I would like to give you a bit of information on this. Okay, so... so the data you have here basically derived from these four major methodologies. Okay, you can have some static data, which is static, solid, ordered information; this is mostly from X-ray crystallography. You can have solution data, dynamic information, and so on... that gives you some special problems, tackling ensembles, conformational ensembles, and different models. And... we will show you some of the tools how to do this. Okay, and recently we are getting more and more structures from cryo-electron microscopy; this is mostly for large assemblies, molecular machines, and this is also some kind of solid information. And also we get information from small-angle X-ray scattering, which is a rough, low-resolution information, giving some data about dimensions, shapes, and interpretive modeling... I mean, structures derived from integrative modeling, let's say someone combines some of these methodologies.

Now the number of structures are incredibly increasing, with an unbelievable pace. Okay, not... the structures that are there are accessible, you can, you can see the statistics here and you can also access it through the PDB. Okay, but... if you see a structure there, you may try to approach the authors to, to get a structure that you need. Now, also it's interesting to see just for a second that the kind of the proportions, the contributions of the different methodologies are changing with the time. You can see, for example, the yellow line, the cryo-EM methodologies; they are extremely... this is kind of most rapidly increasing. So people are more and more interested in large protein machines. Okay, but still on the top, still X-ray crystallography, so the static structures. Okay, and you can see kind of a plateau with the solution structures. Depending on what you want to do, you may want to use different methodologies. Okay, and also nowadays you can go to modeling data in case experimental structures are not available; you can find some of the tools there, although for the modeling we will use other databases—in particular, the AlphaFold—once you understand something about the protein evolution that is kind of incorporated in the AlphaFold methodologies.

PDB Identification Codes

Okay, so this was a very brief introduction to the PDB and now I... together with you, will kind of search different kinds of data in this PDB. So all the PDB structures that are deposited are identified by four-character codes that can be numbers or letters—of course, letters only of the English alphabet. So you can... the first computational exercise is to find out when the identification code ends, based on the statistics of the structures and based on the possible...

(A student asks a question: "Can you repeat that?")

Okay, so I just said that kind of in a... that the first computational exercise is to predict when the code ends, when they terminate, because we have the letters, we have the numbers, and you see the exponential increase in the number of structures... so you can compute in how many years we need to change the identification coding of the PDB structures.

So now let's search for a certain code, okay, related to the problem we are working on today. In this case, I give you the code and this is 1JSU. Okay, and please type it here, like I did, and just enter. Then you can...

Analyzing a Protein Complex (1JSU)

Okay, you get it. So, then you can ask me obviously: "How do I know this code?" Now, there will be a specific, let's say, practical... related to just to finding codes. How to find a structure code, how to find this code and that code... that's kind of a bit complicated thing because you have to find all kinds of codes, but this code is available. If you are super lucky, you can even get it from the Wikipedia; if you read an article and the structure is deposited, it must be present in the article. Okay, but I will show you in a later practical how to collect codes related to a protein, okay? But for the moment, please do accept that we know this code, and the problem we try to study is a problem biologically related to cell cycle regulation, okay? And we are looking at here... it's just the homepage. We are looking at here a complex that basically consists of three proteins, okay: P27, cyclin, and CDK2.

It's not all... I mean, this is just to orient, this is really what I would like to, let's say, analyze. It's not so evident all the time that this information is present, so we need to look at a few things. Again, there is a recap of all these things in a PDB... sorry, in a PDF file, so you can go through later but let's do it now online. So what are you looking at when you get your code? First, you want to make sure that the proteins that are inside are really your proteins, okay? And this you can control in two ways. One way is that immediately you go down kind of to the middle of the page where it says "Macromolecules." And then here, in the Macromolecules section, you have the chance to see... you see here molecule name, which you may not understand... I mean, kinases are molecules that put a phosphate group to their proteins and maybe mediate some kind of signaling, okay? But for the moment, for the computations, you don't need to do this. Okay, but there is a code here, and this code—P24941—will unambiguously identify this protein. We will work later a lot with this code, but this is kind of an unambiguous identifier.

There are many other data that may be relevant to you or may be misleading you; you may discard it at this moment. Then you can go on and you can try to... there's lots of information here which you just discard for the moment. You just go on and you control the number of the chains and the type of the molecules for each chain that they kind of encode. So there's chain B, and this is cyclin A. You can ask me what it means "chain," and chain means that these are covalently bound amino acid residues, and when chain ends, the covalent bond ends as well. Okay, so this means that these are distinct, disjoint segments of protein, and whenever you would like to set up a simulation, you need to deal with them separately. Okay, so we already know about two chains, and then if we go down, we have basically three chains, okay? And each of them, please note, they possess a UniProt number, and this number usually you write down. This is something important for you, we will speak about it a lot.

Heteroatoms and Small Molecules

If we go a little bit further, we may be interested in other issues, which means small molecules. You will see later that they are in a separate section, and each of the lines that contain these small molecules begins with a different kind of identifier. Normally the biomolecules like protein, DNA, or RNA, they start with a simple word "ATOM," where this one starts with the word "HETATM," which means heteroatom. Now why do we care, why do we care about these small molecules? We do care because they may be important for the functionality of the protein; they may be cofactors that may be something that are functionally required. Or here, for example, there is a residue—you see even the chemical formula which may be useful for you—which is a post-translational modification... this is a phosphate group. This is a threonine residue, this part you may recognize, and it's phosphorylated. So this may be somehow related to the function of the protein and you find it in a separate section.

Sequence and Structure Correspondence

So first of all, this is what you need to know: what is in my file, what is that I want to use from my file? There is another issue that you need to look at: if you go to the very top of this page, you find this let's say tab which says "Sequence." Please push it. And you need to look at something else. Okay, the site is a bit slow. So here, it seems that you get something similar information, but let me tell you why we need this. So here you see a list of the chains, and if you click here you can change between the different, let's say, chain identities. But let's just stick with chain A. Obviously, if you want, you can come directly to here and you can read about the macromolecules, but here you cannot read about the heteroatoms or small molecules that may be required for function.

So anyway, what are we doing here? We are double-checking here something. Because there is a sequence of the protein and this is encoded with this number, and Franco will teach you later how to extract this sequence with Biopython. However, if the protein has a region which is disordered—either the C-terminus, N-terminus, or even in between there is a loop or something—then it is possible that the crystallographers, the experimentalists, whatever way they use, they don't see the whole structure. So it is possible that you have a protein of 400 residues, but what is represented in the structure file—we call PDB file, Protein Data Bank file—maybe it's only 300 residues or 200 residues. So one of the first things you need to know is what part of the protein is represented here.

And here you need to come to this line, this kind of lilac or purple line, and you see here something. When you put your cursor on this purple line, you see here "aligned region," and it says UniProt—UniProt is kind of the reference for all the sequences, and when it says UniProt, it should be the reference, the full sequence. And it says position from 1 to 298. And it says for the structure... I can't... you can see AUTH... it's also 1 to 298. So in this case, it's perfect because it's a match, one to one. So we know that the protein is represented almost by 300 residues.

However, it's not always such a simple case; we can check for the others. In many cases, there is a shift. For example here, look at this one. Here we are talking about residues that come between 1 and 260, but in fact, in the UniProt sequence, you see that it's starting at 173. So when you search for identity of residues, this is a crucial information. Later on, I will teach you, or maybe Franco will say a few words about it... this is a huge challenge, to make a correspondence between the full sequence and the PDB file simply because we do, for example, many sequence-based predictions using the full sequence, the original sequence—this we will do next Wednesday, by the way. And at the same time, if you want to, let's say, represent on the PDB or correspond to the PDB structure what we computed, it's very difficult unless we know this correspondence. Okay, so this is something that is really critical. It might sound boring while I'm talking about it, but you may just simply pause all your work for weeks if you cannot resolve this problem. Okay, so this is a very important problem.

Structure Determination Methods and Resolution

Okay, let's go back to the, to the main site, to the "Structure Summary," and let me tell you a few more things that you need to pay attention to. Okay, so when you kind of clarified what is in your structure, okay, you make sure a few other things. One is the method of the structure determination, okay, which you may not know beforehand; in this case, it's X-ray diffraction, so you expect that there is only a single model in this structure, okay? And you may also pay attention to the resolution. Okay, this means, in this case, 2.3 Angstroms. This means that the precision of this measurement is approximately of one C-C bond, a little bit more than that. So they cannot really see anything that goes beyond. Then we don't speak about... this is kind of about the quality of the structure, and R-free values are better the structure is, but depending on your interest, maybe we will have an additional, let's say, extra lecture about structure determination methods where I explain you all these quantities. But anyway, you just, you know, you just have a glance on, on these.

Literature and Cross-References

So afterwards... there is the last thing, you see kind of textual thing, okay, is the thing that is called "Literature." We use mostly as literature the PubMed, okay? This is the NIH site for the articles, and each of the articles has an identifier. Okay, so I will post the article corresponding to this structure, and basically this may be useful pointing out some of the, some of the functional features about the structure itself or about the particular proteins. However, as you are kind of more like machine learning scientists, you will use probably the articles less and you will probably use cross-references to the databases more. Okay, and therefore, again as I forecast, we will have a dedicated practical just to the UniProt and just to cross-linking different kinds of databases, okay, and then you will learn how to do it. We can't do it yet because you don't know several aspects that you wouldn't understand. But I'm forecasting that you will do this kind of cross-linking later.

Downloading the PDB File

So let's derive the data. In principle, what you need to do is to download a so-called legacy PDB file, okay? Please download it, although I downloaded it and I uploaded it for you to Moodle, but please download it. For a second, you may also ask me: "Why there is mmCIF, why there is PDB, and so on?" And there is a bit of information about this in this "Biomolecular Structure Files"... so basically PDB format is a historical format—I will point out some of the problems with this format—and mmCIF is a more standardized, more easy to handle with the computer.

Exploring the PDB File Header

I will display now a PDB file and briefly tell you what you find in the PDB file itself. So the PDB file, again, is a historical format. Okay, and in the beginning, it's all storytelling. There are many things that you may be interested in, okay, which you can also get from the structure article. What things you may be interested in? Well, here immediately it says "Chain B has a fragment this one," so this kind of things, the identification you may be interested in. The other thing you may be interested in: from which organism your molecule is derived. And here your organism is Homo sapiens, but it may be expressed in, it may be produced for the structural analysis in a different system... and you could see here that expression system, it is this organism. It may be important because, for example, if you use a human protein or any of the eukaryotic protein, and then you would have an important phosphorylation site, for example, you would lose all these... you would lose all these in the final structure because in the bacteria there is no phosphorylation. In fact, you can see here that it was expressed in the E. coli.

However, when you work, you normally just ignore the header. The header cause lots of headache for computational scientists, I would say for decades, and nowadays it's done in a very different way, okay? So for the moment, in fact, we will just discard the header, okay? And we arrive at the information... there may be just one thing that you may use: sometimes they report, if it's correct, a remark on missing residues. Okay, and this will come up later if they couldn't determine a position of a few residues... then you should know what to do with these. This may also appear as disorder in the original PDB file.

The ATOM Records

But what you really want—and I'm sure you want to compute as soon as possible—what you really want is really this part of the file, and for most of the problems, in fact, you can just remove the header, you don't need it. So what you see here, this is the classical PDB format. As I said, all the macromolecules, biomolecules start with "ATOM." This simply means that it's a biomolecule. Then there is a numbering, which is a consecutive numbering, and it doesn't... it's kind of ignorant if something is missing from the structure and so on, it's just an arbitrary numbering.

And then, this is the crucial part, this is the classification part, okay, the following four columns. Okay, let's take in a bit different order than it is in the file. So, in column four, you find a letter that is not necessarily A or B or C—it depends on the authors—but that's the identity, that's the chain identifier. So that's important. That refers to the particular macromolecule that we checked. Then you have the residue number, which is in column five. And this one refers basically to the PDB residue. So this is a number in the structure; this may not be the number, as I said, in the reference sequence file that you have just seen. In the chain B, it starts at 170-something. You must make a note of this, and later on if you want to make some correspondence, it's becoming very important.

Then you have the residue identifier, which is a three-letter code here; good for you, easier to read. However, there are problems here... so you are dealing with experimental structures. You want to see, obviously, you accept: "Okay, maybe my sequence is shifting by, I don't know, 170-something residues. Okay, that's fine, but then I have a full correspondence." Unfortunately, this is not the case. Because there are some long residues, long side chains, that may be very mobile. And in that case, researchers may not see the end of it. And very unfortunate... and we will see some of the long residues, for example the positively charged residues, the lysine or the arginine. And unfortunately, if they don't see the last residues, the last groups, the atoms... by the way, only the heavy atoms, because we are talking about crystal structures...

Speaker 1: ...I preferred check to- to run a, let's say, a cross-check in reference to the reference sequence file to understand what we are dealing with. Whether there are parts that are missing, maybe they just identified the first half of the residue but did not identify the- the second part, thinking we are dealing with experimental data.

Also, it is possible that these numbers, which are the residue numbers again listed in column five, it is possible that the experimentalists list as consecutive numbers whereas it's possible that there is a ten-residue gap here. So, unfortunately, the very first part of dealing with the structures starts with some kind of counting, some- some kind of administration to properly assign what is what, where do I have the gaps, if it's really continuous or not.

Okay, and then in column three, these are the atom names, and these are standard names. When I was teaching to you the residue amino acids and then, you know, the primary sequence and the- the grammar of protein sequences, in one of the first slides, which- which we intentionally skipped, I- I put all the Greek letters corresponding to the different atoms.

Okay, so these are the- these are the Greek letters. I mean, OG1 is basically O-gamma-1, okay, transcribed to Latin letters, but these are standard. And you do need to pay attention to this when you, for example, would like to start molecular simulation and you would like to assign each of these atoms to a potential energy surface that we will do later, okay?

I'm taking this long because unless you really understand this, you won't be able to transmit this information to, let's say, a molecular simulation file and generate potential energy surface. Whatever you have after it's- it's very easy because you have the X, Y, and Z coordinates, that's fine.

And in the last two columns, you have the following information: before the last—well, the very last column is actually the atom type, but most of the time it's trivial. But I'm talking about kind of the last columns, the one and there is a large number. Now, the one before the last column is almost—almost always one, unless the structure has undergone a very special experimental method or anomalous refinement, when they present kind of two conformers for each of the atoms. It's possible and then the one become half and half, so this basically tells you the occupancy.

But you don't have much to do with this for the moment, it's—but in the last column, there is the first information that is kind of a physical information or physics-based information. So that tells you the mobility. In principle, the very last number would be related—it's called the B-factor.

This is related to the thermal motion of the atoms. Normally, when they do the refinement, when they kind of a little bit adjust the experimental data with the help of some very primitive simulation methodology, they normally use isotropic refinement assuming that isotropic motion of the- of the atoms. If they are performing more sophisticated methods or high resolution, then you may find some secondary lines with different numbers talking about anisotropic refinement and motions to- to different directions.

Now, what is the meaning of this? We will kind of represent these B-factors later. So this tells you something about the mobility, but it's not just strictly the mobility. Don't forget these are experimental evaluations of experimental data, so this is also kind of taking up some of the errors. When you find—when you see these numbers like here, I don't know, like in the 20s, 50s, it's all very fine. However, when you start seeing these numbers high—80, 90, sometimes it's even 100 and touching the previous column—it means that there is a problem there, there is too large mobility there, so that's kind of a suspicious part of the structure that you may have a look at.

So let's very quickly go—normally I don't edit files like this, but I'm still on the PDB. So if I- if I finish this A chain... okay, I was here. So if I finish this A chain, okay, each of the chains ends with a TER tag, okay? This means that the chain ends and you find a specific atom name OXT. Okay, then you go through all the other chains, okay, in a similar manner you- you have all the correspondence. And then when you finish all this macromolecule section, then you would go—then you see it's terminated, chain C, and then you start this special section which I said it's HETATM and here you find all the molecules, cofactors, whatever.

The structure is- is similar, the structure of the representation, and here also you find all the so-called crystallographic water molecules. So these are the water molecules captured in- in the structure itself and they are basically have kind of fixed position. This is a bit of exaggeration here because normally these water molecules have much higher B-factors, so this is also slightly suspicious.

2.2 Tutorial: PyMol visualization of the 1JSU protein (PDB entry)

If you want to create a new selection group for a specific chain (e.g., Chain A):

```
select my_chain, chain A
```

If you want to change the actual name of some object in the menu:

```
set_name old_name, new_name
```

Now, the philosophy of PyMOL is that you show everything until you don't say hide. So you know, you can display several options and then for particular subsets, whatever, you can just say hide, okay? Until you don't say hide, it will be all on the screen, okay?

The label option which is here is the least used, let's say, because it's not let's say it's not so good for publication quality, but the it may orient you in one way or another so temporarily you can- you can label your residues, for example, three letters or one letter or whatever. And color, this is what you use probably most frequently because you want to distinguish between the different parts of your protein.

So I'm trying to accelerate now. I have this structure and well we said it contains three molecules but have no idea. If I color, normally I look for, you know, where the enter, where the molecule starts, when does it end. So I- I put—I click on color and there are options like spectrum and I just choose rainbow. If I choose this rainbow something, what it will do for me, it will color very dark in the very beginning of the molecule which is here until red which is the- which is the very C-terminus.

However, again, I have three molecules here. How do I- how do I know? Well, there are two ways. One is that I remember from the PDB that I had and I can type this command for example, sel chain A, just very simply. And this will be chain A, okay, and you see here it selected all, it replaced how many atoms are selected. And either in the action, I can- I can rename it. It's good to rename it and I don't rename it as chain A, can rename but I- I remember it was CDK2, this was a molecule name and that's useful to rename it as the molecule name. So I rename it as CDK2.

I can do the same, then I- you need to click out because also until you don't click out, everything is kind of added. So you need to click out of this. And you can also select in the same way, let's say, select chain B, but you can—I put into the PDF file there is—it's selection alphabet area, you can choose residues from this to this, only atom C alpha, so I mean it's a real programmable language, okay? I- I'm just showing you the very first baby steps once you feel it you can program it.

So you can select chain B and you can also change the naming in the command line, you can say set name set—first when select it's called sel and change it to cyclin. And then I have another subset which is called cyclin. At this stage it's useful to save the session, to- at the top you can say file, save session as, and maybe update class practical or whatever, and from then on don't- don't forget saving.

Okay, there is a third way how to select the subsets and this may be useful. So if you—maybe in your screen is not the right bottom but you find the letter which is called S. And this S letter corresponds to the sequence. So if I click on it, I should get a long series of letters with all the sequence here and if I observe it carefully, it's listed here A, and then if I scroll it somewhere it's listed B, and so on.

So even I could do my selections here if I didn't want to, you know, use command line or whatever. There are lots of—I mean, these one-letter code, even if you don't—if you have not learned them but you know they signify amino acids. Then have lots of O O O, this means they are water molecules.

Okay, we will undisplay water molecules because for the moment we- we don't want them to be seen and you can just say here all hide and you look for waters. And I undisplay them, so it's, you know, cleaner. Later on, I can display some of these water.

So anyway, we can go to chain C here and I can scroll it and with the shift bar I can go until the end of it and this will be my chain C, the third molecule, which was called in the PDB file p27. So that's great. I have now, I save my session, I have now the three molecules, but it's not a very meaningful coloring.

I should see—but I prefer seeing these three molecules as—with different colors based on the subunits, let's do this first. So does every—does everyone have the- the subsets? Is everything okay? Franco? Is everything okay? No questions?

Speaker 2: No questions now.

Speaker 1: Okay, perfect. So then you can come here CDK2 color and you choose whatever color you like. I will tell you why I choose grey. I will just choose grey 70.

And then for the other, for the cyclin, I will also choose grey but I choose a darker one, maybe grey 40. And at this point it's very useful to change the background color, which you can change here display or the very—so this is the very top menu, you have several options for- for the display and here you can change the background and I will just change it to white.

Okay, and I turn it a little bit. There is another feature which people who have some problems with vision may utilize. There is something like depth cue or fogging. If you switch it off, you- you have less 3D dimensionality but just as a—just as a display but you have stronger colors. I usually use it.

You see it's—so this is actually quite nice. You can adjust this display setting. For example, this you can see the helices, you can see some beta strands and some of these loops are very thin. So there are all kinds of options to change it. For example, I can say set cartoon loop radius, okay, and I give a number, maybe 0.4 or something. Ah, I- even I made a mistake. Okay, I hope it's okay... yes, it accepted.

So you see that everything has grown. But you can change everything, you can change the look of the helix, you can change separately for the beta strands, so you can, you know, you can play around with it. So in this manner, for example, I save—by the way I save the session. Okay, I can start seeing how this is constructed from three molecules. Two are kind

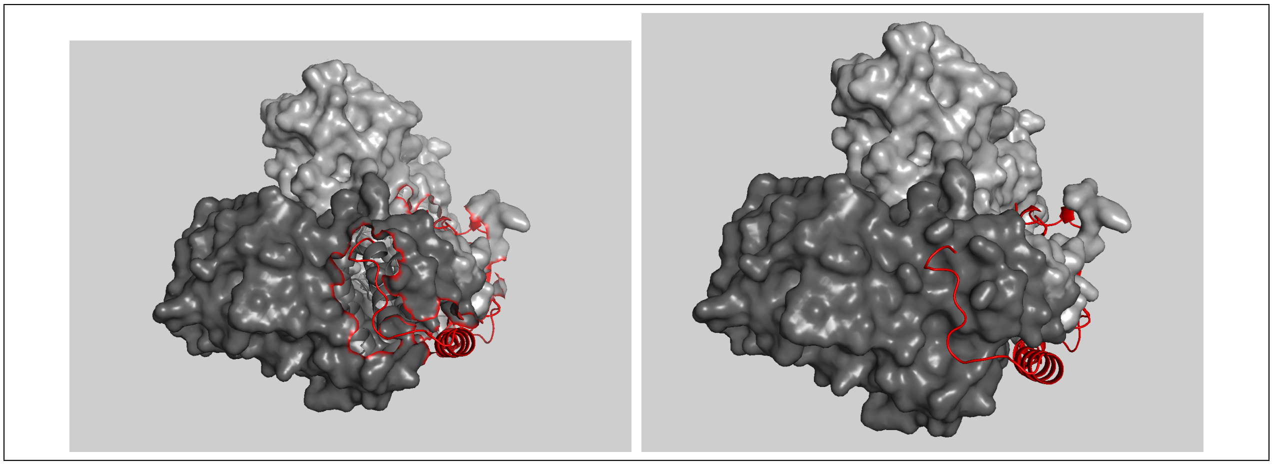


Figure 2.1: Caption

of making the unit and one is kind of around.

Now what is around is an inhibitor. So let's do as follows. Let's display this whole thing as a surface and let's- let's see the overall organization. So we go to the S button here in the- in command and we say show surface.

And this is quite nice. First of all, you can see how rugged the protein surface is and you can see the kind of a hug that this little red gives to the other two proteins. But this kind of a hug inhibits cell cycle to go on and when this kind of hug is- is removed in one way or another, then the cell cycle continues. So this is an important hug.

Now this is very nice and I suggest, you know, you can send it to your beloved ones on 14th of February, but this is not so useful if you want to do computational analysis. Because in principle, you want to see how the hug occurs, which residues interact with which other residues, okay?

And for that matter, it would be useful to- to undisplay the- the surface for the p27, the red one. So I would hide the surface, okay, just for this one, and then see how it interacts with the others. But there is still a problem here—of course there are always problems, but we create and resolve them. So you can see that there are some holes, right, on the surface of the other two molecules. Obviously because it would be the place for the surface of p27.

So to solve this kind of problems, I posted another structure. Basically what you need to use is to generate the structure which contains only chain A and chain B, so only the two grey ones, okay? And then you basically take a- you- you read that in. I'm just showing to you now, but you can repeat it at home.

So you can open my file that I posted for you and this is—I'm sorry, uh... yes, so it is called CDK2 cyclin. So this is without the p27. I read in that file, you can see already there is another file. If I hide for the second the- the surface, you should see that I have a file somewhere, maybe it's centered in another part. And basically I can just say—and all of these are listed in your PDF files—align this CDK2 cyclin on the 1GSU.

And once you do this, this is a rigid body alignment, so rigid body rotation and translation. You see already that this kind of cyan color appeared and you see here that the root mean square deviation is zero. Because it's zero because it's basically the same structure. Unfortunately to make a very nice image, I would quickly need to do the same thing, which instead of typing I would just do here. So I need to—I need to define the two sets of the CDK2 here... okay.

So I need to define in order to generate the same picture, I will just call now chain A and chain B, but you can call in whatever way you want. Okay... okay. Okay. I color these in the same way how it was colored before. Okay.

And I undisplay everything, the CDK2 and the cyclin from the previous structure. I undisplay, hide, hide, and I also hide everywhere the waters. So in that manner now I have a composite structure. And if I say now to show the surface for this cyclin structure, which doesn't know in principle about the p27, you can see that this is your first nice image. Okay, I also posted the images in your Moodle so that you can play around with them. This is the first place when you can start feeling how this is anchored here, how this is anchored here and so on.

So let me tell you now in a few words what are the next following steps that you can do. First to save your session. Now if you are pleased with the image, you can just go to this ray or draw and you can adjust to your DPI and you can just say ray. And you can save the image and this image will be saved in a manner—okay, I imitate here—will be saved in a manner that will be completely transparent in the background.

Speaker 2: We cannot see the settings for the background ray.

Speaker 1: You cannot see. Then in the academic maybe you cannot do this, but anyway if you save the image you should be able to ray it and you should be able to get the image with a transparent background. And when you read it into any kind of program, Word or PowerPoint or whatever, I mean you can all put something on the background. When I save the image, whatever name, I just call it class image. Okay, once I saved it, if you click on it, it becomes to your original image and so on.

So this is kind of just the very beginning of the beautiful story. So let me tell you now in a few words what comes after it and you will continue now with Franco who- who can show you how to really compute with it. But you need to be able to tackle it. So in a few minutes, I- I tell you what else you can do.

So this is a very nice story of this cell cycle inhibition and there are some competitors here, there are some viruses which can imitate, so there is a third PDF file in your Moodle section that tells you the whole story, that tells you many other structures, how it comes, how one replaces another, how you search because the next step would be how you search for the motifs that are contacting each other. Then you can, you know, see whether it's in the mobile part, not mobile part, is hydrophobic, not hydrophobic.

So the play starts from here. There are many, many things that you can do—identifying residues that are in contact, or for example you try to show—sorry—you try to show for example the B-factors, which are the mobile part, which are the fixed part. You can see that it's relatively rigid when it touches and so on. So you can do all this kind of games and they are all in the PDF files.

The second step that you can do and we will do with Franco with BioPython that you find three structures...